

● MEDIUM

● ALGEBRA

● ANSWER KEY

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# Algebra

10 answers · Rationale-rich · Teach from doc

# 15

**Q1****B****B.**  $\frac{4}{17}$ 

Choice B is correct. The second equation in the given system is  $y = 9x + 1$ . Substituting  $9x + 1$  for  $y$  in the first equation in the given system yields  $8x + 9x + 1 = 5$ , which is equivalent to  $17x + 1 = 5$ . Subtracting 1 from both sides of this equation yields  $17x = 4$ . Dividing both sides of this equation by 17 yields  $x = \frac{4}{17}$ .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

**Q2****A****A.** The average number of trees per hectare in the park

Choice A is correct. It's given that a neighborhood consists of a 2-hectare park and a 35-hectare residential area and that the total number of trees in the neighborhood is 3,934. It's also given that the equation  $2x + 35y = 3,934$  represents this situation. Since the total number of trees for a given area can be determined by taking the number of hectares times the average number of trees per hectare, this must mean that the terms  $2x$  and  $35y$  correspond to the number of trees in the park and in the residential area, respectively. Since  $2x$  corresponds to the number of trees in the park, and 2 is the size of the park, in hectares,  $x$  must represent the average number of trees per hectare in the park.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

**Q3**

**B**

B. 3.04

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Choice B is correct. It's given that the function  $f(x) = 55.20 - 0.16x$  gives the estimated surface water temperature, in degrees Celsius, of a body of water on the  $x$ th day of the year.

Substituting 326 for  $x$  in the given function yields  $f(326) = 55.20 - 0.16(326)$ , which is equivalent to  $f(326) = 55.20 - 52.16$ , or  $f(326) = 3.04$ . Therefore, the estimated surface water temperature, in degrees Celsius, of this body of water on the 326th day of the year is 3.04.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the rate of change, in degrees Celsius per day, of the estimated surface water temperature.

Choice D is incorrect. This is the change, in degrees Celsius, in the estimated surface water temperature over 326 days.

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**Q4****C****C. 12**

Choice C is correct. Let  $x$  represent the number of children in a whale-watching tour group. Let  $y$  represent the number of adults in this group. Because it's given that 21 people are in a group and the group consists of adults and children, it must be true that  $x + y = 21$ . Since the company's revenue is 60 dollars per child, the total revenue from  $x$  children in this group was  $60x$  dollars. Since the company's revenue is 80 dollars per adult, the total revenue from  $y$  adults in this group was  $80y$  dollars. Because it's given that the total revenue for this group was 1,440 dollars, it must be true that  $60x + 80y = 1,440$ . The equations  $x + y = 21$  and  $60x + 80y = 1,440$  form a linear system of equations that can be solved to find the value of  $x$ , which represents the number of children in the group, using the elimination method. Multiplying both sides of the equation  $x + y = 21$  by 80 yields  $80x + 80y = 1,680$ . Subtracting  $60x + 80y = 1,440$  from  $80x + 80y = 1,680$  yields  $(80x + 80y) - (60x + 80y) = 1,680 - 1,440$ , which is equivalent to  $80x - 60x + 80y - 80y = 240$ , or  $20x = 240$ . Dividing both sides of this equation by 20 yields  $x = 12$ . Therefore, 12 people in the group were children.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the number of adults in the group, not the number of children in the group.

Choice D is incorrect and may result from conceptual or calculation errors.

**Q5****1.636**

Accept also: 18/11

The correct answer is  $\frac{18}{11}$ . For a line that passes through the points  $x_1, y_1$  and  $x_2, y_2$  in the  $xy$ -plane, the slope of the line can be calculated using the slope formula,  $m = \frac{y_2 - y_1}{x_2 - x_1}$ . It's given that a line passes through the points 4, 6 and 15, 24 in the  $xy$ -plane. Substituting 4, 6 for  $x_1, y_1$  and 15, 24 for  $x_2, y_2$  in the slope formula,  $m = \frac{y_2 - y_1}{x_2 - x_1}$ , yields  $m = \frac{24 - 6}{15 - 4}$ , or  $m = \frac{18}{11}$ . Therefore, the slope of the line is  $\frac{18}{11}$ . Note that 18/11 and 1.636 are examples of ways to enter a correct answer.

**Q6****D**

D.  $B = 212 - \frac{x}{550}$

Choice D is correct. It's given that the boiling point of water at sea level is 212°F and that for every 550 feet above sea level, the boiling point of water is lowered by about 1°F. Therefore, the change in the boiling point of water  $x$  feet above sea level is represented by the expression

$-\frac{x}{550}$ . Adding this expression to the boiling point of water at sea level gives the equation for

the boiling point  $B$  of water, in °F,  $x$  feet above sea level:  $B = -\frac{x}{550} + 212$ , or

$$B = 212 - \frac{x}{550}.$$

Choices A and B are incorrect and may result from using the boiling point of water at sea level as the rate of change and the rate of change as the initial boiling point of water at sea level.

Choice C is incorrect and may result from representing the change in the boiling point of water as an increase rather than a decrease.

**Q7****A**

A. -4, -8

Choice A is correct. The given system of linear equations can be solved by the substitution method. The first equation in the given system of equations defines  $y$  as  $6x + 16$ . Substituting  $6x + 16$  for  $y$  in the second equation of the given system of equations yields  $-7x - (6x + 16) = 36$ . Applying the distributive property on the left-hand side of this equation yields  $-7x - 6x - 16 = 36$ , or  $-13x - 16 = 36$ . Adding 16 to both sides of this equation yields  $-13x = 52$ . Dividing both sides of this equation by -13 yields  $x = -4$ . Substituting -4 for  $x$  in the first equation of the given system of equations,  $y = 6x + 16$ , yields  $y = 6(-4) + 16$ , or  $y = -8$ . Therefore, the solution  $(x, y)$  to the given system of equations is  $(-4, -8)$ .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

**Q8****A**

A.  $y = 5x + 6$

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Choice A is correct. The linear relationship between  $x$  and  $y$  can be represented by the equation  $y = mx + b$ , where  $m$  is the slope of the line in the  $xy$ -plane that represents the relationship, and  $b$  is the  $y$ -coordinate of the  $y$ -intercept. The slope can be computed using any two points on the line. The slope of a line between any two points,  $x_1, y_1$  and  $x_2, y_2$ , on the line can be calculated using the slope formula,  $m = \frac{y_2 - y_1}{x_2 - x_1}$ . In the given table, each value of  $x$  and its corresponding value of  $y$  can be represented by a point  $x, y$ . In the given table, when the value of  $x$  is 1, the corresponding value of  $y$  is 11 and when the value of  $x$  is 2, the corresponding value of  $y$  is 16. Therefore, the points 1, 11 and 2, 16 are on the line. Substituting 1, 11 and 2, 16 for  $x_1, y_1$  and  $x_2, y_2$ , respectively, in the slope formula yields  $m = \frac{16 - 11}{2 - 1}$ , or  $m = 5$ . Substituting 5 for  $m$  in the equation  $y = mx + b$  yields  $y = 5x + b$ . Substituting the first value of  $x$  in the table, 1, and its corresponding value of  $y$ , 11, for  $x$  and  $y$ , respectively, in this equation yields  $11 = 5(1) + b$ , or  $11 = b + 5$ . Subtracting 5 from both sides of this equation yields  $6 = b$ . Substituting 6 for  $b$  in the equation  $y = 5x + b$  yields  $y = 5x + 6$ . Therefore, the equation  $y = 5x + 6$  represents the linear relationship between  $x$  and  $y$ .

Choice B is incorrect. For this relationship, when the value of  $x$  is 1, the corresponding value of  $y$  is 16, not 11.

Choice C is incorrect. For this relationship, when the value of  $x$  is 2, the corresponding value of  $y$  is 17, not 16.

Choice D is incorrect. For this relationship, when the value of  $x$  is 1, the corresponding value of  $y$  is 17, not 11.

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The correct answer is 257. It's given that a linear model estimates the population of a city from 1991 to 2015. Since the population can be estimated using a linear model, it follows that there is a constant rate of change for the model. It's also given that the model estimates the population was 57 thousand in 1991, 224 thousand in 2011, and  $x$  thousand in 2015. The change in the population between 2011 and 1991 is  $224 - 57$ , or 167, thousand. The change in the number of years between 2011 and 1991 is  $2011 - 1991$ , or 20, years. Dividing 167 by 20 gives  $167 / 20$ , or 8.35, thousand per year. Thus, the change in population per year from 1991 to 2015 estimated by the model is 8.35 thousand. The change in the number of years between 2015 and 2011 is  $2015 - 2011$ , or 4, years. Multiplying the change in population per year by the change in number of years yields the increase in population from 2011 to 2015 estimated by the model:  $8.35 \times 4$ , or 33.4, thousand. Adding the change in population from 2011 to 2015 estimated by the model to the estimated population in 2011 yields the estimated population in 2015. Thus, the estimated population in 2015 is  $33.4 + 224$ , or 257.4, thousand. Therefore to the nearest whole number, the value of  $x$  is 257.

## Q10

C

C.  $y = 16x + 3$   
 $y = 16x + 19$

Choice C is correct. A system of two linear equations in two variables,  $x$  and  $y$ , has no solution if the graphs of the lines represented by the equations in the  $xy$ -plane are distinct and parallel. The graphs of two lines in the  $xy$ -plane represented by equations in slope-intercept form,  $y = mx + b$ , where  $m$  and  $b$  are constants, are parallel if their slopes,  $m$ , are the same and are distinct if their  $y$ -coordinates of the  $y$ -intercepts,  $b$ , are different. In the equations  $y = 16x + 3$  and  $y = 16x + 19$ , the values of  $m$  are each 16, and the values of  $b$  are 3 and 19, respectively. Since the slopes of these lines are the same, and the  $y$ -coordinates of the  $y$ -intercepts are different, it follows that the system of linear equations in choice C has no solution.

Choice A is incorrect. The lines represented by the equations in this system are a vertical line and a horizontal line. Therefore, this system has a solution, 3, 5, rather than no solution.

Choice B is incorrect. The two lines represented by these equations have different slopes and the same  $y$ -coordinate of the  $y$ -intercept. Therefore, this system has a solution, 0, 6, rather than no solution.

Choice D is incorrect. The two lines represented by these equations are a horizontal line and a line with a slope of 5 that have the same  $y$ -coordinate of the  $y$ -intercept. Therefore, this system has a solution, 0, 5, rather than no solution.

### Notes

Accept equivalent forms (e.g.  $0.25 \equiv 1/4$ ). For grid-in items, decimal and fraction answers are both valid.

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